

# INDU 211 · Comprehensive Topic Guide (Part 4)

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## Chapter 5: Material Handling, Distribution & Routing

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*Built on the teacher's Lecture 5.0 slides, with depth from the course textbook (Hicks, Chapter 5: Material Handling, Distribution, and Routing).*

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### 1. Why Material Handling Matters

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Material handling means **using equipment to move materials** inside and between facilities. It is **non-productive (non-value-added)**: moving a part does not change it, yet it costs money, time and space. Chapter 4 already told us that material handling is **30% to 95% of production cost**, which is why layout and handling are designed together.

## Material Handling

- To Use Equipment to Move Materials Internally
  - Non-productive (non-value added)

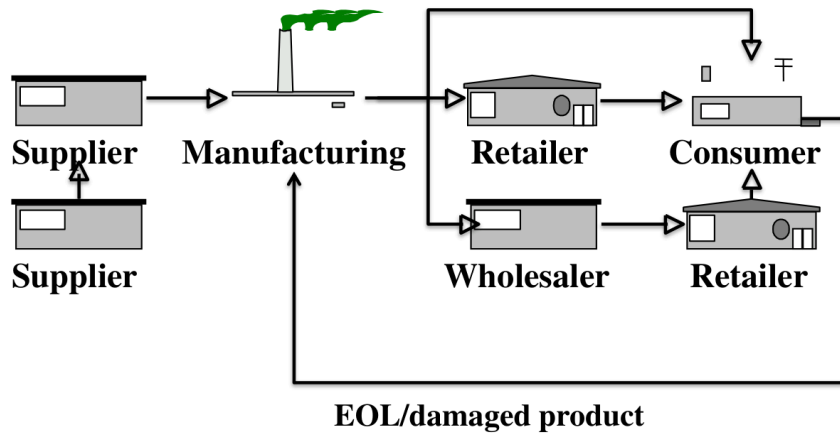


Figure 1: Material handling, Lecture 5.0, slide 3.

**Reading the figure:** material moves from **suppliers** to **manufacturing**, then to a **wholesaler**, a **retailer** and finally the **consumer**. The arrow returning from the consumer is the reverse flow of **end-of-life or damaged product**. Every arrow is a handling and transport step that adds cost but no value.

## Material Handling

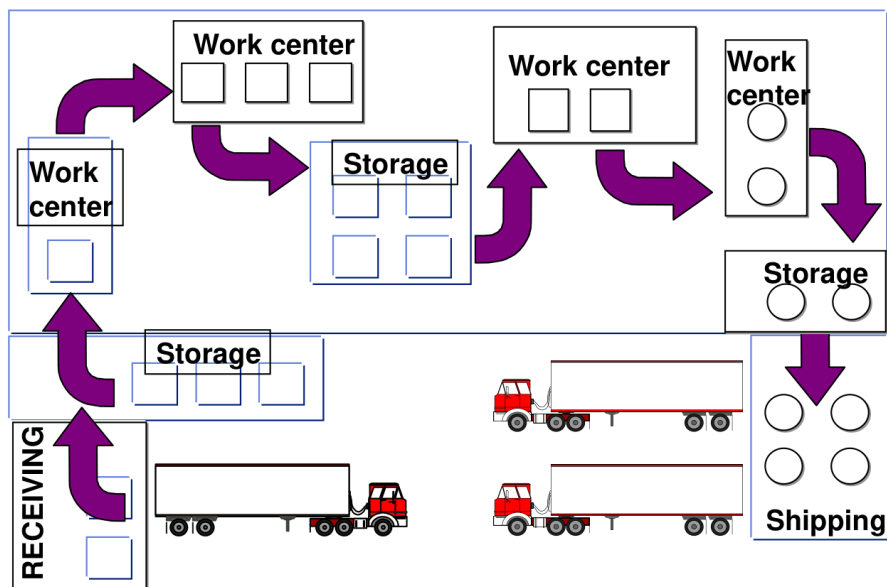


Figure 2: Material handling within a facility, Lecture 5.0, slide 4.

**Reading the figure:** inside the plant, material arrives at **receiving**, goes to **storage**, travels between **work centres** and intermediate storage, and leaves through **shipping** onto trucks. The purple arrows are the internal moves an industrial engineer tries to shorten, combine or eliminate.

**From the textbook (Turner et al., §5.1):** material movement occurs  
*any time*

a product or part moves from one place to another, and it usually accounts for a large share of the cost of producing the product. Reducing the number and length of moves is one of the cheapest ways to cut cost.

## 2. Material Handling Equipment

| Equipment                                      | What it does                                                                        | Typical use                    |
|------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------|
| <b>Conveyors</b>                               | Move homogeneous material from one <b>fixed point</b> to another at a constant rate | Assembly lines, parcel sorting |
| <b>Industrial trucks</b> (forklifts)           | Move loads along <b>varying paths</b> , intermittently                              | Job shops, warehouses          |
| <b>Cranes and hoists</b>                       | Overhead lifting of heavy loads                                                     | Steel mills, heavy assembly    |
| <b>Containers and racks</b>                    | Store and handle bulk material; better use of space                                 | Pallets, bins, carts           |
| <b>Elevators and lifts</b>                     | Vertically raise or lower material; fixed location                                  | Multi-floor plants             |
| <b>Automatic guided vehicles (AGV)</b>         | Driverless vehicles on predetermined paths                                          | Automated plants, hospitals    |
| <b>AS/RS</b> (automated storage and retrieval) | Storage racks + computer control + crane                                            | High-density warehouses        |

## Material handling Equipment

- Conveyors
  - From one fixed point to another
  - Homogeneous material supplied at a constant rate
- Industrial trucks
  - Varying paths
  - Intermittent load
  - Job shops



Figure 3: Conveyors and industrial trucks, Lecture 5.0, slide 5.

**Reading the slide:** the roller conveyors (top right) fix the path, which suits steady, identical loads. The forklifts (bottom) can go anywhere with intermittent loads, which suits job shops where every order takes a different route.

**The choice of equipment follows the layout:** product layouts use conveyors, process layouts use trucks.

## Material handling Equipment

- Automatic guided vehicles (AGV)
  - Driverless vehicle
  - Predetermined paths

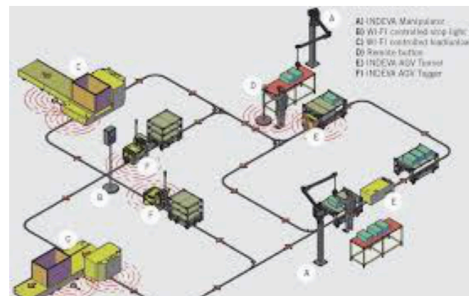


Figure 4: Automatic guided vehicles, Lecture 5.0, slide 8.

**Reading the slide:** AGVs are driverless carts that follow predetermined paths (wires, tape or laser navigation). The layout sketch (bottom right) shows the guide paths connecting stations: flexible like trucks, but automated like conveyors.

## Material handling Equipment

- Automated storage and retrieval system (AS/RS)
  - Storage rack systems
  - Control system (computer)
  - Crane



*Figure 5: Automated storage and retrieval system (AS/RS), Lecture 5.0, slide 9.*

**Reading the photos:** tall racks with narrow aisles and a computer-controlled crane that stores and retrieves loads. AS/RS trades a large investment for **space utilisation** (building up, not out) and fast, accurate picking.

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### 3. Principles of Material Handling

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#### Principles of materials handling

- Planning (planned not evolved)
  - Systems (material movement cycle)
  - Material flow (smooth)
  - Simplification (motion economy)
  - Gravity (gravity feed bins, roller conveyors)
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- Space utilization (high racks with narrow aisle stacking trucks in warehouses)



Figure 6: Principles of materials handling, Lecture 5.0, slide 10 (continued on slide 11).

| Principle                  | Meaning                                                       | Example                                          |
|----------------------------|---------------------------------------------------------------|--------------------------------------------------|
| <b>Planning</b>            | Handling is planned, not left to evolve                       | Handling plan drawn with the layout              |
| <b>Systems</b>             | Treat material movement as a cycle from receiving to shipping | Integrated receiving–storage–production–shipping |
| <b>Material flow</b>       | Keep flow smooth                                              | Avoid back-tracking and cross-traffic            |
| <b>Simplification</b>      | Motion economy: eliminate, combine, simplify moves            | Deliver straight to the point of use             |
| <b>Gravity</b>             | Use gravity when possible                                     | Gravity-feed bins, roller conveyors on a slope   |
| <b>Space utilisation</b>   | Use the cube, not just the floor                              | High racks with narrow-aisle stacking trucks     |
| <b>Unit size</b>           | Move the largest practical accumulated load                   | Pallets, containers                              |
| <b>Automation</b>          | Use powered and automated equipment where it pays             | Powered conveyors, automatic pallet stackers     |
| <b>Equipment selection</b> | Choose based on all aspects of the material and move          | Weight, shape, frequency, distance               |
| <b>Standardisation</b>     | Standard equipment and load sizes                             | Standard pallets and containers                  |
| <b>Adaptability</b>        | Equipment that can do a variety of tasks                      | Variable-speed conveyors                         |
| <b>Maintenance</b>         | Plan preventive and emergency maintenance                     | Scheduled conveyor servicing                     |
| <b>Safety</b>              | Protect the operator                                          | Guarding, safe stacking heights                  |

**Exam tip:** questions often give an example and ask which principle it illustrates. Gravity-feed bins and roller conveyors → **gravity**; pallets → **unit size**; high racks → **space utilisation**; "planned, not evolved" → **planning**.

## 4. Quantitative Material Handling Models (Turner §5.3)

While the lecture slides introduce material handling qualitatively, the course textbook (

*Turner, Mize, Case & Nazemetz*

, §5.3) establishes the rigorous quantitative mathematical models that industrial engineers use to size fleets of handling equipment (such as AGVs) and analyze conveyor capacity.

### A. Automated Guided Vehicle (AGV) Fleet Sizing Model

To determine how many automated guided vehicles ( $N_v$ ) are required to support a production schedule without starving workstations, we balance the **total system workload** against the **available operating time per vehicle**.

## 1. Delivery Cycle Time ( $T_c$ )

The time required for an AGV to complete one full pickup-and-delivery cycle consists of four distinct components:

$$T_c = T_{\text{load}} + \frac{L_d}{v_c} + T_{\text{unload}} + \frac{L_e}{v_c}$$

Where:

- $T_c$  = Total delivery cycle time (min/delivery)
- $T_{\text{load}}$  = Time required to load the vehicle at the pickup station (min)
- $L_d$  = Travel distance loaded from pickup to destination (m or ft)
- $v_c$  = Vehicle travel speed (m/min or ft/min)
- $T_{\text{unload}}$  = Time required to unload the vehicle at the dropoff station (min)
- $L_e$  = Travel distance empty from previous dropoff to next pickup (m or ft)

## 2. Hourly Workload ( $WL$ )

If the facility generates delivery requests across multiple delivery loops  $i = 1, 2, \dots, m$ , with demand rate  $D_i$  (deliveries per hour) on loop  $i$ :

$$WL = \sum_{i=1}^m D_i \times T_{ci} \quad (\text{minutes per hour})$$

## 3. Available Vehicle Time & Traffic Factor ( $T_{\text{avail}}$ )

In an industrial environment, an AGV cannot operate for 100% of the 60 minutes in an hour due to battery recharging, blocking at intersections, queueing behind other vehicles, and unexpected breakdowns. The net available time per vehicle per hour is:

$$T_{\text{avail}} = 60 \times A \times TF \quad (\text{min/hr per vehicle})$$

Where:

- $A$  = **Vehicle Availability** (operational uptime proportion, e.g., 0.90 to 0.98)
- $TF$  = **Traffic Factor** ( $0 < TF \leq 1.0$ ), accounting for delays at guide-path intersections, passing conflicts, and waiting for personnel. Typical values range from 0.85 to 0.95.

## 4. Fleet Sizing Formula

The theoretical number of AGVs is the workload divided by available time. Because a fractional vehicle cannot be deployed, we take the ceiling (round up to the next whole integer):

$$N_v = \left\lceil \frac{WL}{T_{\text{avail}}} \right\rceil = \left\lceil \frac{\sum D_i T_{ci}}{60 \times A \times TF} \right\rceil$$

**Worked Numerical Example (Turner §5.3):** An automated assembly plant requires  $D = 24$  pallet transfers per hour between receiving and the main assembly line.

- Loaded travel distance:  $L_d = 120$  m
- Empty return travel distance:  $L_e = 90$  m
- AGV travel speed:  $v_c = 40$  m/min
- Pickup loading time:  $T_{\text{load}} = 1.25$  min
- Dropoff unloading time:  $T_{\text{unload}} = 1.00$  min
- System availability:  $A = 0.95$ , Traffic congestion factor:  $TF = 0.90$

**Step 1: Calculate cycle time per delivery:**

$$T_c = 1.25 + \frac{120}{40} + 1.00 + \frac{90}{40} = 1.25 + 3.00 + 1.00 + 2.25 = 7.50 \text{ min/delivery}$$

**Step 2: Calculate total system workload per hour:**

$$WL = 24 \text{ deliveries/hr} \times 7.50 \text{ min/delivery} = 180.0 \text{ min/hr}$$

**Step 3: Calculate available operating time per vehicle:**

$$T_{\text{avail}} = 60 \times 0.95 \times 0.90 = 51.30 \text{ min/hr per AGV}$$

**Step 4: Determine fleet size:**

$$N_v = \left\lceil \frac{180.0}{51.30} \right\rceil = \lceil 3.509 \rceil = 4 \text{ AGVs}$$

*Conclusion:*

Exactly 4 automated guided vehicles must be purchased and scheduled. 3 vehicles would provide only  $3 \times 51.3 = 153.9$  minutes of capacity, failing to meet the required 180 min/hr workload.

## B. Conveyor System Analysis & Delivery Rates

For fixed-path continuous conveyors (roller, belt, or overhead monorail carriers), the textbook defines two core operating metrics:

**1. Carrier Delivery Rate ( $R_d$ ):**

$$R_d = \frac{v_c}{s_c} \quad (\text{carriers per minute})$$

where  $v_c$  is conveyor line speed (m/min) and  $s_c$  is the center-to-center spacing between carriers or hooks (m).

**2. Total Parts Delivery Rate ( $R_f$ ):** If each carrier holds  $n_p$  parts:

$$R_f = n_p \times R_d = n_p \frac{v_c}{s_c} \quad (\text{parts per minute})$$

3. **Station Passage / Clearance Time ( $T_p$ ):** The time a carrier of length  $L_c$  spends passing a fixed station point:

$$T_p = \frac{L_c}{v_c}$$

This sets the strict upper bound on manual loading/unloading without stopping the conveyor line.

## 5. Routing: The Travelling Salesman Problem (TSP)

**Problem:** one vehicle leaves a depot (plant A), visits every location exactly once, and returns, so that the **total distance is minimised**. Example: routing a forklift through a warehouse, or one truck supplying all warehouses.

### Material routing

- Travelling salesman problem (TSP)
  - Routing a fork truck in a warehouse
- Example 1- the truck must start from the plant (A), visit all warehouses, and return to the plant such that the total travelled distance is minimized.
- One truck can supply all warehouses

| To<br>From | A  | B  | C  | D  | E  | F  | G  |
|------------|----|----|----|----|----|----|----|
| A          | -  | 14 | 21 | 20 | 6  | 24 | 9  |
| B          | 14 | -  | 10 | 9  | 9  | 10 | 11 |
| C          | 21 | 10 | -  | 1  | 15 | 9  | 21 |
| D          | 20 | 9  | 1  | -  | 14 | 9  | 20 |
| E          | 6  | 9  | 15 | 14 | -  | 19 | 9  |
| F          | 24 | 10 | 9  | 9  | 19 | -  | 21 |
| G          | 9  | 11 | 21 | 20 | 9  | 21 | -  |

Figure 7: Material routing (TSP example 1), Lecture 5.0, slide 12.

**Reading the matrix:** row = from, column = to. The matrix is symmetric ( $A \rightarrow B = B \rightarrow A = 14$ ), and the diagonal is blank (no trip from a place to itself).

### Why a heuristic?

The TSP is **very difficult to solve optimally** when there are many stops: with  $n$  stops there are  $(n - 1)!/2$  different tours. For 7 locations that is  $6!/2 = 360$  tours, which could be enumerated; for 20 it is about  $6 \times 10^{16}$ , which cannot. So we use **heuristics**: fast rules that give a good, feasible route that is not guaranteed optimal.

### Nearest Neighbour method (Lecture 5.0, slide 14)

**Rule:** from the current location, always go to the **closest unvisited** location; when all are visited, return to the start.

| Step | From | Closest unvisited           | Distance | Running total |
|------|------|-----------------------------|----------|---------------|
| 1    | A    | E                           | 6        | 6             |
| 2    | E    | B (tie with G at 9; take B) | 9        | 15            |
| 3    | B    | D                           | 9        | 24            |
| 4    | D    | C                           | 1        | 25            |
| 5    | C    | F                           | 9        | 34            |
| 6    | F    | G (only one left)           | 21       | 55            |
| 7    | G    | back to A                   | 9        | <b>64</b>     |

The optimal tour is **A-G-B-F-C-D-E-A = 60**. Nearest neighbour is  $\frac{64-60}{60} = 6.7\%$  worse, and if the tie at E is broken toward G the route becomes 69. **Heuristics can be sensitive to tie-breaking**: the greedy early choices force expensive late legs (F→G = 21).

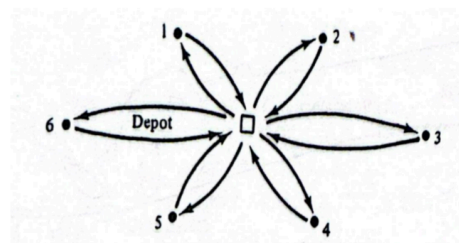
## 6. Vehicle Routing and the Clark-Wright Savings Method

When one truck cannot carry all the demand, we need **several routes**: the **vehicle routing problem (VRP)**, also called the multiple travelling salesman problem. Each route starts and ends at the depot, and the load on each route must not exceed the **truck capacity**.

**Lecture example 2**: supply 6 warehouses (B–G) from depot A; truck capacity 25,000 units; demands B 5,000, C 7,000, D 10,000, E 4,000, F 6,000, G 10,000 (total 42,000, so at least  $\lceil 42,000/25,000 \rceil = 2$  trucks).

### Clark-Wright procedure

- Initial route generation
- Calculate the savings as the result of pairing nodes
- Rank the savings in a descending order
- Start with the first combination to add to the route and continue on the list until no more stop can be added
  - Truck capacity
  - Route feasibility
  - All destinations are visited



$$S_{ij} = C_{oi} + C_{oj} - C_{ij}$$

$C_{oi}$  = distance from depot to stop  $i$   
 $C_{oj}$  = distance from depot to stop  $j$   
 $C_{ij}$  = distance from stop  $i$  to stop  $j$

Figure 8: Clark-Wright procedure, Lecture 5.0, slide 19.

**Reading the diagram:** the "flower" on the right is the starting solution, with one out-and-back trip from the depot to every stop. Joining stops  $i$  and  $j$  into one trip removes the two returns to the depot and adds the direct leg  $i \rightarrow j$ . The distance saved is

$$S_{ij} = C_{0i} + C_{0j} - C_{ij}$$

where  $C_{0i}$  is the depot-to- $i$  distance and  $C_{ij}$  the distance between the two stops.

### The procedure (slide 19)

1. **Initial routes:** one route per stop.
2. **Calculate savings**  $S_{ij}$  for every pair.
3. **Rank** the savings in descending order.
4. **Link** pairs in that order, as long as the truck capacity holds, the route stays feasible (a stop can only be joined at the end of a route), and all destinations are eventually visited.

**Worked savings:**  $S_{CD} = 21 + 20 - 1 = 40$  (the largest),  $S_{CF} = 21 + 24 - 9 = 36$ ,  $S_{DF} = 20 + 24 - 9 = 35$ .

### Clark-Wright procedure

Ranked combinations:

| <i>Stops</i> | <i>Saving</i> | <i>Stops</i> | <i>Saving</i> | <i>Stops</i> | <i>Saving</i> |
|--------------|---------------|--------------|---------------|--------------|---------------|
| 1 CD         | 40            | 6 BD         | 25            | 11 BE        | 11            |
| 2 CF         | 36            | 7 BG         | 12            | 12 EF        | 11            |
| 3 DF         | 35            | 8 CE         | 12            | 13 CG        | 9             |
| 4 BF         | 28            | 9 DE         | 12            | 14 DG        | 9             |
| 5 BC         | 25            | 10 FG        | 12            | 15 EG        | 6             |

Figure 9: Ranked combinations, Lecture 5.0, slide 20.

## Building the routes

| Rank | Pair | Saving | Decision                                      | Route load |
|------|------|--------|-----------------------------------------------|------------|
| 1    | C–D  | 40     | Start route Depot–C–D–Depot                   | 17,000     |
| 2    | C–F  | 36     | Add F at the C end: Depot–F–C–D–Depot         | 23,000     |
| 3    | D–F  | 35     | Both already on the same route: skip          | —          |
| 4    | B–F  | 28     | Adding B makes 28,000 > 25,000: <b>reject</b> | —          |
| ...  | ...  | ...    | Continue; build the second route              | ...        |
|      |      |        | Second route Depot–E–B–G–Depot                | 19,000     |

**Final routes:** Depot–F–C–D–Depot ( $24 + 9 + 1 + 20 = 54$ ) and Depot–E–B–G–Depot ( $6 + 9 + 11 + 9 = 35$ ), total **89** ( $\times 100 = 8,900$  miles).

**From the textbook (Turner et al., §5.3.2–5.3.3):** the same savings logic is used for public-sector routing (garbage collection, school buses, mail). Real VRPs add time windows and driver-hour limits; exact solutions are hard, so heuristics like Clark-Wright and computer software are standard practice (Lecture 5.0, slide 22).

## 7. Exam Checklist

- [ ] Material handling is **non-value-added** and 30–95% of production cost.
- [ ] Match equipment to layout: conveyors (fixed path) vs trucks (varying path) vs AGV vs AS/RS.
- [ ] Name the principle from an example (gravity, unit size, space utilisation, planning...).
- [ ] AGV Fleet Sizing:  $T_c = T_L + L_d/v_c + T_U + L_e/v_c$ ;  $N_v = \lceil WL/(60 \cdot A \cdot TF) \rceil$ .
- [ ] Conveyor Delivery Rate:  $R_d = v_c/s_c$ ; flow rate  $R_f = n_p R_d$ .
- [ ] Nearest neighbour: always the closest **unvisited** stop; add the return leg; state the tie-breaking rule.
- [ ] Savings:  $S_{ij} = C_{0i} + C_{0j} - C_{ij}$ ; rank descending; check **capacity** before every link.
- [ ] Minimum number of trucks = total demand  $\div$  capacity, **rounded up**.